

KL Divergence, VAE, and ELBO

From surprise and extra bits to the Gaussian closed form and ELBO

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Step-by-step lecture slides

Roadmap

- 01 Surprise, entropy, and extra bits
- 02 Closed-form Gaussian KL from mean & variance
- 03 Why a VAE uses this KL
- 04 ELBO, step by step
- 05 Training: reconstruction + KL

What is KL divergence?

Two distributions P (nature) and Q (model) on the same space:

$$\text{KL}(P\|Q) = \sum_x P(x) \log \frac{P(x)}{Q(x)} \quad \text{KL}(P\|Q) = \int p(x) \log \frac{p(x)}{q(x)} dx$$

Non-negative

≥ 0 , and $= 0$ iff $P = Q$

Asymmetric

$\text{KL}(P\|Q) \neq \text{KL}(Q\|P)$

Surprise gap

Large where P lives and Q is near 0

Read $\text{KL}(P\|Q)$ as: extra cost of using Q when samples actually come from P .

Entropy + cross-entropy

$$\text{KL}(P\|Q) = H(P, Q) - H(P)$$

Entropy $H(P)$

$$H(P) = - \sum P(x) \log P(x)$$

Average surprise of P itself. Optimal code length.

Cross-entropy $H(P, Q)$

$$H(P, Q) = - \sum P(x) \log Q(x)$$

Code length when we encode P 's data with Q .

$$\text{KL}(P\|Q) = H(P, Q) - H(P) = \text{extra bits wasted}$$

Surprise — a weather forecast

$$I(x) = \log \frac{1}{P(x)} = -\log P(x)$$

Independent events multiply; logs make surprise additive.

Fair forecast

sun 50%, rain 50%

$$I(\text{sun}) = I(\text{rain}) = \log_2 2 = \mathbf{1 \text{ bit}}$$

Sunny forecast

sun 90%, rain 10%

$$I(\text{rain}) = \log_2 10 \approx \mathbf{3.32 \text{ bits}}$$

Rain is far more surprising.

If rain is certain ($P = 1$), then $I = 0$: no surprise, no information.

Entropy — average surprise

$$H(P) = \mathbb{E}_{x \sim P}[I(x)] = - \sum_x P(x) \log P(x)$$

Fair coin

$H = 1$ bit — one yes/no question.

Certain rain

$H = 0$ — no uncertainty.

Three-way $P = (0.5, 0.25, 0.25)$

sun / cloud / rain

$$\begin{aligned} H(P) &= 0.5 \cdot 1 + 0.25 \cdot 2 + 0.25 \cdot 2 \\ &= \mathbf{1.5} \text{ bits} \end{aligned}$$

A code matched to P uses a short label for sun, longer labels for cloud and rain.

Cross-entropy vs KL — extra bits

True weather $P = (0.5, 0.25, 0.25)$. Wrong model $Q = (0.25, 0.5, 0.25)$ (thinks cloud is most likely).

Entropy

1.50

bits, true P

Cross-entropy

1.75

bits with Q 's code

KL wasted

0.25

bits = $1.75 - 1.50$

$$H(P, Q) = -0.5 \log_2 0.25 - 0.25 \log_2 0.5 - 0.25 \log_2 0.25 = 1.75$$

P is nature. Q is the model. KL is the extra cost of trusting Q .

Discrete calculation (nats)

$$P = (0.7, 0.2, 0.1), \quad Q = (0.5, 0.3, 0.2) \quad (\log = \ln)$$

Term	$\log(P_i/Q_i)$	$\times P_i$	Contribution
$i = 1$	0.3365	0.7	+0.2356
$i = 2$	-0.4055	0.2	-0.0811
$i = 3$	-0.6931	0.1	-0.0693
Sum = $\text{KL}(P\ Q)$			0.0852 nats

Positive $\Rightarrow Q$ is slightly off. Same recipe as weather, different log base.

Forward vs reverse KL

$\text{KL}(P\|Q) \neq \text{KL}(Q\|P)$. The first argument is the reference.

Forward $\text{KL}(P\|Q)$

Mode covering

Punishes Q for missing mass of P . A single Gaussian Q **spreads** to cover both peaks of a bimodal P .

Reverse $\text{KL}(Q\|P)$

Mode seeking

Punishes Q for mass where P is near 0. Q **sits on one peak** and ignores the rest.

VAE uses $\text{KL}(q_\phi(\mathbf{z} \mid \mathbf{x}) \parallel p(\mathbf{z}))$: the encoder must not wander where the prior forbids.

Setup — two 1-D Gaussians

$$q(z) = \mathcal{N}(\mu, \sigma^2) \quad p(z) = \mathcal{N}(0, 1)$$

$$q(z) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(z - \mu)^2}{2\sigma^2}\right), \quad p(z) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{z^2}{2}\right)$$

Why a closed form?

A VAE evaluates this KL on every batch. We want μ and σ^2 only — no leftover integral.

Closed form — Step 1: log densities

$$\log q(z) = -\frac{1}{2} \log(2\pi\sigma^2) - \frac{(z - \mu)^2}{2\sigma^2}$$

$$\log p(z) = -\frac{1}{2} \log(2\pi) - \frac{z^2}{2}$$

$$\text{KL}(q\|p) = \mathbb{E}_{z \sim q} [\log q(z) - \log p(z)]$$

After cancellation of $\log 2\pi$:

$$\text{KL}(q\|p) = \mathbb{E}_q \left[-\frac{1}{2} \log \sigma^2 - \frac{(z - \mu)^2}{2\sigma^2} + \frac{z^2}{2} \right]$$

Closed form — Step 2: moments of q

Under $z \sim \mathcal{N}(\mu, \sigma^2)$:

$$\mathbb{E}[z] = \mu$$

$$\mathbb{E}[(z - \mu)^2] = \sigma^2$$

$$\mathbb{E}[z^2] = \sigma^2 + \mu^2$$

$$\begin{aligned}\text{KL} &= -\frac{1}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \underbrace{\mathbb{E}[(z - \mu)^2]}_{\sigma^2} + \frac{1}{2} \underbrace{\mathbb{E}[z^2]}_{\sigma^2 + \mu^2} \\ &= -\frac{1}{2} \log \sigma^2 - \frac{1}{2} + \frac{1}{2}(\sigma^2 + \mu^2)\end{aligned}$$

Closed form — Step 3: mean and variance

$$\text{KL}(\mathcal{N}(\mu, \sigma^2) \parallel \mathcal{N}(0, 1)) = \frac{1}{2}(\mu^2 + \sigma^2 - 1 - \log \sigma^2)$$

$$\mu^2$$

Penalty for moving the latent mean off 0

$$\sigma^2 - \log \sigma^2$$

Penalty for variance $\neq 1$

$$-1$$

Offset so $\mu = 0, \sigma^2 = 1$ gives $\text{KL} = 0$

Check. $\mu = 0, \sigma^2 = 1 \Rightarrow \frac{1}{2}(0 + 1 - 1 - 0) = 0$. Encoder matches the prior.

General Gaussians and d dimensions

$$\text{KL}(\mathcal{N}(\mu_1, \sigma_1^2) \parallel \mathcal{N}(\mu_2, \sigma_2^2)) = \log \frac{\sigma_2}{\sigma_1} + \frac{\sigma_1^2 + (\mu_1 - \mu_2)^2}{2\sigma_2^2} - \frac{1}{2}$$

VAE prior $\mu_2 = 0$, $\sigma_2 = 1$ recovers the previous box.

Diagonal d -D latents (standard VAE): KL **adds** over coordinates

$$\text{KL} = \frac{1}{2} \sum_{j=1}^d (\mu_j^2 + \sigma_j^2 - 1 - \log \sigma_j^2)$$

Worked Gaussian examples

Mean shift only

$$q = \mathcal{N}(0.2, 1), p = \mathcal{N}(0, 1)$$

$$\text{KL} = \frac{1}{2}(0.04 + 1 - 1 - 0) \\ = \mathbf{0.02}$$

Variance matches the prior.

Too-narrow encoder

$$q = \mathcal{N}(0, 0.25), p = \mathcal{N}(0, 1)$$

$$\text{KL} = \frac{1}{2}(0.25 - 1 - \log 0.25) \\ \approx \mathbf{0.318}$$

Tiny σ is expensive under the prior.

Pick Gaussians so the analytic KL is known — then you can sanity-check any sampler.

Closed form vs Monte Carlo

$$\text{KL}(q\|p) = \mathbb{E}_{z \sim q}[\log q(z) - \log p(z)]$$

Monte Carlo

Draw $z^{(i)} \sim q$, average log-ratios.

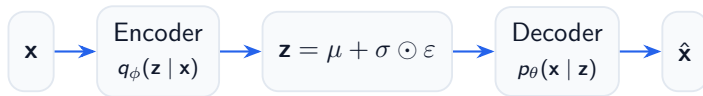
Unbiased but high variance; estimates can even go **negative**.

VAE choice

Plug μ, σ^2 into the closed form.

Cheap, exact (given μ, σ), no leftover sampling noise.

Why a VAE needs KL



Without KL

Isolated tiny codes. Reconstruction works; samples from $\mathcal{N}(0, I)$ do not.

With KL

q_{ϕ} stays near $p(\mathbf{z}) = \mathcal{N}(0, I)$. Generation from the prior is meaningful.

Target — maximize data likelihood

$$\log p_{\theta}(\mathbf{x}) = \log \int p_{\theta}(\mathbf{x} \mid \mathbf{z}) p(\mathbf{z}) d\mathbf{z}$$

Problem

This integral is intractable for neural decoders (we cannot sum over all $\mathbf{z}$).

Idea. Introduce $q_{\phi}(\mathbf{z} \mid \mathbf{x})$ and derive a **tractable lower bound**: the ELBO.

ELBO — Step 1: rewrite with q_ϕ

Insert $\int q_\phi(\mathbf{z} \mid \mathbf{x}) d\mathbf{z} = 1$ and use $p(\mathbf{x}) = p(\mathbf{x}, \mathbf{z})/p(\mathbf{z} \mid \mathbf{x})$:

$$\log p_\theta(\mathbf{x}) = \mathbb{E}_{q_\phi(\mathbf{z} \mid \mathbf{x})} \left[\log \frac{p_\theta(\mathbf{x}, \mathbf{z})}{p_\theta(\mathbf{z} \mid \mathbf{x})} \right]$$

Meaning

Still **exact**. No approximation yet — only an expectation under the encoder.

ELBO — Step 2: split, expose KL

$$\log p_{\theta}(\mathbf{x}) = \mathbb{E}_{q_{\phi}} \left[\log \frac{p_{\theta}(\mathbf{x}, \mathbf{z})}{q_{\phi}(\mathbf{z} | \mathbf{x})} \right] + \text{KL}(q_{\phi}(\mathbf{z} | \mathbf{x}) \| p_{\theta}(\mathbf{z} | \mathbf{x}))$$

$$\log p_{\theta}(\mathbf{x}) = \text{ELBO}(\theta, \phi; \mathbf{x}) + \underbrace{\text{KL}(q_{\phi} \| p_{\theta}(\mathbf{z} | \mathbf{x}))}_{\geq 0}$$

The gap is how far the encoder is from the true posterior.

ELBO — Step 3: the bound

$$\log p_{\theta}(\mathbf{x}) \geq \text{ELBO}(\theta, \phi; \mathbf{x})$$

Floor

ELBO sits under the true log-likelihood

Train

Raising ELBO raises a bound on $\log p(\mathbf{x})$

Tight

Equality iff $q_{\phi} = p_{\theta}(\mathbf{z} \mid \mathbf{x})$

ELBO — Step 4: VAE training form

$$\text{ELBO} = \underbrace{\mathbb{E}_{q_\phi}[\log p_\theta(\mathbf{x} \mid \mathbf{z})]}_{\text{reconstruction}} - \underbrace{\text{KL}(q_\phi(\mathbf{z} \mid \mathbf{x}) \parallel p(\mathbf{z}))}_{\text{KL regularizer}}$$

Reconstruction

Can the decoder rebuild $\mathbf{x}$ from $\mathbf{z}$?
Often $-\text{MSE}$ or $-\text{BCE}$.

KL regularizer

Does the encoder stay near $\mathcal{N}(0, I)$?
Use the closed form we derived.

Practical training loss

We **minimize** $-\text{ELBO}$:

$$\mathcal{L} = \mathcal{L}_{\text{recon}} + \beta \cdot \text{KL}(q_{\phi}(\mathbf{z} \mid \mathbf{x}) \parallel \mathcal{N}(\mathbf{0}, \mathbf{I}))$$

$$\beta = 1$$

Standard VAE

$$\beta > 1$$

Stronger structure

$$\beta \rightarrow 0$$

Plain autoencoder

Reparameterization

Sampling $\mathbf{z} \sim q_\phi$ is stochastic. Push noise into ε :

$$\varepsilon \sim \mathcal{N}(0, I), \quad \mathbf{z} = \boldsymbol{\mu}_\phi(\mathbf{x}) + \boldsymbol{\sigma}_\phi(\mathbf{x}) \odot \varepsilon$$

ε

No parameters

$\boldsymbol{\mu}, \boldsymbol{\sigma}$

Deterministic in $\mathbf{x}$

Gradients

Flow into encoder ϕ

One training step

1. Encode $\mathbf{x} \rightarrow (\boldsymbol{\mu}, \boldsymbol{\sigma})$
2. Sample $\varepsilon \sim \mathcal{N}(0, I)$; $\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \varepsilon$
3. Decode $\mathbf{z} \rightarrow \hat{\mathbf{x}}$; compute $\mathcal{L}_{\text{recon}}$
4. $\text{KL} = \frac{1}{2} \sum_j (\mu_j^2 + \sigma_j^2 - 1 - \log \sigma_j^2)$
5. $\mathcal{L} = \mathcal{L}_{\text{recon}} + \beta \text{KL}$; backprop; update θ, ϕ

Numerical ELBO — 1-D latent

One point: $\mu = 0.5$, $\sigma^2 = 0.8$, $\mathcal{L}_{\text{recon}} = 0.40$

KL

$$\begin{aligned} & \frac{1}{2}(0.25 + 0.8 - 1 - \log 0.8) \\ &= \frac{1}{2}(0.27314) \approx \mathbf{0.1366} \end{aligned}$$

Loss / ELBO

$$\begin{aligned} \mathcal{L} &\approx 0.40 + 0.1366 = \mathbf{0.5366} \\ \text{ELBO} &\approx -0.5366 \end{aligned}$$

Higher ELBO = tighter (better) bound on $\log p(\mathbf{x})$.

Takeaways

1. KL = extra bits of using model Q when nature is P . Always ≥ 0 .
2. Closed form: $\frac{1}{2}(\mu^2 + \sigma^2 - 1 - \log \sigma^2)$ from $\mathbb{E}[(z - \mu)^2] = \sigma^2$ and $\mathbb{E}[z^2] = \sigma^2 + \mu^2$.
3. VAE uses that KL so latents stay near $\mathcal{N}(0, I)$.
4. $\log p(\mathbf{x}) = \text{ELBO} + \text{KL}(q\|p(\mathbf{z} \mid \mathbf{x}))$.
5. Train: reconstruction quality minus KL; reparameterize $\mathbf{z}$.

Practice. Show $\text{KL} = 0$ when $\mu = \mathbf{0}$, $\sigma = \mathbf{1}$. What does that say about the encoder?